Chapter 17: Inference about a Population Mean

The Practice of Statistics in the Life Sciences
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Part IV: Specific Inference Procedures

Why Chapter 17 Matters

Chapter 14 introduced confidence intervals and hypothesis tests for a population mean when the population standard deviation σ was known.

In practice, σ is rarely known. Chapter 17 replaces it with the sample standard deviation s and develops procedures based on the t distributions.

Main Question

How can a sample be used to estimate or test a population mean when both μ and σ are unknown?

The interpretation of confidence intervals and P-values remains the same. The probability model used for the calculations changes.

Chapter Roadmap

The chapter develops six connected ideas.

1. Conditions for inference about a population mean
2. The standard error of a sample mean
3. The family of t distributions
4. One-sample t confidence intervals and tests
5. Matched pairs t procedures
6. Robustness when the population is not exactly Normal

The theory comes first. Each procedure is then applied to a life-science example and reproduced in R.

Known and Unknown Population Variation

The difference between the procedures in Chapters 14 and 17 concerns the population standard deviation.

	Known σ	**Unknown** σ
Standard error	$\frac{\sigma}{\sqrt{n}}$	$\frac{s}{\sqrt{n}}$
Standardized statistic	z	t
Reference distribution	Standard Normal	t with $n-1$ degrees of freedom
Typical use	Introductory model	Most applications

Estimating σ adds uncertainty. The heavier tails of the t distribution account for that extra uncertainty.

The Inference Process

A one-sample analysis should keep the calculation connected to the research question.

1. **State:** Identify the population and the population mean μ
2. **Plan:** Select a one-sample or matched pairs t procedure
3. **Solve:** Check the design and distributional conditions, then calculate
4. **Conclude:** Interpret the interval or test in the study context

Design Before Formula

A correct t calculation cannot remove bias from an inappropriate sample or support a causal conclusion that the study design does not justify.

Conditions for Inference about a Mean

One-sample t procedures rely on two main conditions.

- **Randomness and independence:** The observations can be treated as a simple random sample from a much larger population, or as outcomes from another defensible random process
- **Shape:** The population distribution is Normal, or the sample is large enough for the sampling distribution of $\bar{x}$ to be approximately Normal

The random-sampling condition determines whether the sample can represent the target population. The shape condition determines whether the t distribution gives reliable critical values and P-values.

Random Sampling and Independence

The sample should represent the population named in the conclusion.

- A simple random sample gives every possible sample of size n an equal chance of selection
- If sampling occurs without replacement, the population should be much larger than the sample
- A common guideline is $n \leq 0.10N$, where N is the population size
- Repeated measurements from the same individual are not independent observations

Scope of Inference

When a convenience sample is used, calculations may still summarize the observed data, but generalization to a broader population requires caution.

Normality and Sample Size

The t procedures are exact when the population distribution is Normal. They are often approximately reliable for non-Normal populations when the sample is sufficiently large.

- With a small sample, inspect the data for skewness and outliers
- As n increases, the Central Limit Theorem improves the Normal approximation for $\bar{x}$
- Outliers remain important because both $\bar{x}$ and s are sensitive to extreme values
- A large sample does not make a poorly defined population or biased sample appropriate

Always examine a graph of the observations before applying a t procedure.

Standard Deviation and Standard Error

These quantities describe different kinds of variation.

Quantity	**Formula**	**What it describes**
Sample standard deviation	s	Variation among individual observations in one sample
Standard error of $\bar{x}$	$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$	Estimated variation among sample means from repeated samples of size n

The standard error describes the precision of the sample mean as an estimate of μ.

The Standard Error of the Mean

When σ is unknown, the sample standard deviation estimates it.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \qquad \longrightarrow \qquad SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

- A smaller s produces a smaller standard error
- A larger n produces a smaller standard error
- Multiplying n by 4 divides the standard error by 2
- The value of s changes from sample to sample, so the estimated standard error also changes

This sample-to-sample variation in s is why the Standard Normal distribution no longer gives the correct reference model.

From the z Statistic to the t Statistic

If σ is known, standardization gives

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

If σ is unknown, replace it with s:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Both statistics measure the distance between the sample mean and a population value in standard-error units. The t statistic has more variability because its denominator is estimated from the same sample.

The One-Sample t Statistic

Suppose a simple random sample of size n comes from a Normal population with unknown mean μ and unknown standard deviation σ.

One-Sample t Statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

This statistic follows a t distribution with

$$df = n - 1$$

degrees of freedom.

The statistic tells us how many estimated standard errors the sample mean lies above or below μ.

Why a New Distribution Is Needed

The Standard Normal model treats the denominator $\sigma/\sqrt{n}$ as fixed.

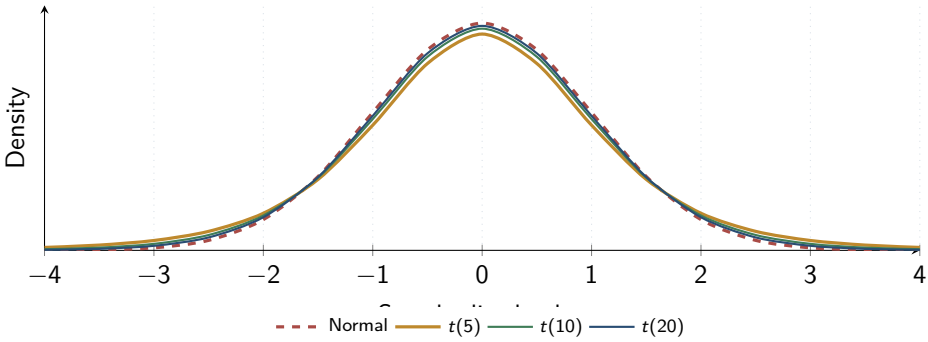
The t statistic uses $s/\sqrt{n}$. Because s changes from sample to sample, the standardized statistic varies more than a z statistic.

Effect on Inference

The t distributions place more probability in the tails than the Standard Normal distribution. Confidence intervals are therefore wider, and a given standardized distance produces a larger P-value.

As the sample size increases, s estimates σ more accurately and the difference between t and z becomes smaller.

The Family of t Distributions



Properties of the t Distributions

Every t density curve is

- Symmetric about 0
- Single-peaked and bell-shaped
- More spread out than the Standard Normal curve
- Identified by its degrees of freedom

Smaller degrees of freedom produce heavier tails. As the degrees of freedom increase, the t distribution approaches the Standard Normal distribution.

$$t(df) \longrightarrow N(0,1) \quad \text{as } df \text{ increases}$$

Degrees of Freedom

For one-sample inference about a mean,

$$df = n - 1$$

The sample deviations from the sample mean must satisfy

$$\sum_{i=1}^{n} (x_i - \bar{x}) = 0$$

After $n - 1$ deviations are known, the final deviation is determined by this constraint. The calculation therefore has $n - 1$ independent pieces of information available for estimating variation.

Practical Use

The degrees of freedom select the appropriate t curve for critical values and tail probabilities.

Normal and t Critical Values

The heavier tails of a t distribution lead to larger critical values.

Reference distribution	Degrees of freedom	95% critical value
t	5	2.571
t	10	2.228
t	20	2.086
t	50	2.009
Standard Normal	∞	1.960

The difference is most noticeable for small samples. It becomes modest as the degrees of freedom increase.

Student's t and the Guinness Brewery

William S. Gosset developed the t distribution while working for the Guinness brewery in the early twentieth century.

His work addressed problems with small samples, including the selection of barley and hops. Guinness restricted employees from publishing company research, so Gosset published under the name "Student."

Statistical Importance

The method accounts for the uncertainty created when a small sample is used to estimate an unknown population standard deviation.

The name **Student's t distribution** recognizes Gosset's published work.

Areas and Critical Values in R

R uses `pt()` for cumulative probabilities and `qt()` for quantiles.

```
# Area to the left of t = -3 under a t distribution with 10 df
pt(q = -3, df = 10)
# [1] 0.006671828

# Point with 0.95 to its left and 0.05 to its right
qt(p = 0.95, df = 10)
# [1] 1.812461
```

- `pt()` answers an area question
- `qt()` answers a cutoff question
- Both functions require the degrees of freedom

Example 17.1: Reading t Probabilities

For a t distribution with 10 degrees of freedom:

- The area to the left of $t = -3$ is approximately 0.0067
- The point with probability 0.05 to its right is approximately 1.812

$$P(T \leq -3) = 0.0067$$

$$P(T \geq 1.812) = 0.05$$

Because the distribution is symmetric,

$$P(T \leq -1.812) = P(T \geq 1.812) = 0.05$$

The One-Sample t Confidence Interval

For a random sample of size n from a population with unknown mean μ and unknown standard deviation σ, a level C confidence interval is

Confidence Interval for μ

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

where t^* is the critical value from the $t(n-1)$ distribution with central area C.

The margin of error is

$$m = t^* \frac{s}{\sqrt{n}}$$

The interval is exact for a Normal population and approximately valid for sufficiently large random samples from non-Normal populations.

The Four-Step Confidence-Interval Process

1. **State:** Define the population and the mean μ to be estimated
2. **Plan:** Choose a confidence level and a one-sample t interval
3. **Solve:** Check the sampling and shape conditions, then calculate $\bar{x}$, s, SE, t^* and the interval
4. **Conclude:** Interpret the interval for the population named in the study

Interpretation

A level C procedure captures the true population mean in proportion C of repeated random samples under the stated conditions.

Example 17.2: Testosterone in Adolescent Males

A study recorded plasma testosterone concentrations for 25 obese adolescent males. The measurements are in nanomoles per litre of blood.

0.30	0.24	0.19	0.17	0.18	0.23	0.24	0.06	0.15
0.17	0.18	0.17	0.15	0.12	0.25	0.25	0.25	0.32
0.35	0.37	0.39	0.46	0.49	0.42	0.36		

Question

What is a 95% confidence interval for the mean plasma testosterone concentration among obese adolescent males similar to those represented by the study?

Example 17.2: Parameter and Plan

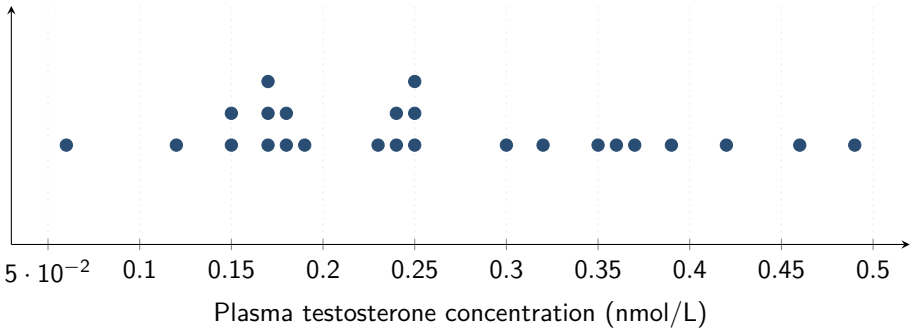
Let μ denote the population mean plasma testosterone concentration for obese adolescent males represented by the study.

The study aims to estimate μ, and the population standard deviation is unknown. The planned procedure is therefore a one-sample t confidence interval.

Important Scope Limitation

The participants came from a medical centre rather than a simple random sample of all obese adolescent males. The strongest conclusion applies to a population similar to the individuals who participated.

Example 17.2: Distribution of the Data



The dotplot shows no extreme outliers and no severe skewness. A one-sample t interval is reasonable for these 25 observations.

Example 17.2: Summary Statistics

The sample statistics are

$$n = 25 \qquad \bar{x} = 0.2584 \text{ nmol/L} \qquad s = 0.1115 \text{ nmol/L}$$

The estimated standard error is

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{0.1115}{\sqrt{25}} = 0.0223 \text{ nmol/L}$$

This standard error estimates how sample means would vary across repeated samples of 25 comparable participants.

Example 17.2: Critical Value and Margin of Error

The degrees of freedom are

$$df = n - 1 = 25 - 1 = 24$$

For a 95% interval, the two tails contain 0.025 each, so

$$t^*_{24} = 2.064$$

The margin of error is

$$m = t^* \frac{s}{\sqrt{n}} = 2.064(0.0223) \approx 0.0460 \text{ nmol/L}$$

Example 17.2: Confidence Interval

$$\bar{x} \pm m = 0.2584 \pm 0.0460$$

$$(0.2124,\ 0.3044)\ \text{nmol/L}$$

Conclusion

We are 95% confident that the population mean plasma testosterone concentration for obese adolescent males similar to those studied lies between approximately 0.212 and 0.304 nmol/L.

The confidence level describes the long-run performance of the interval-producing method.

Example 17.2: Limits of the Interval

The interval quantifies random sampling uncertainty under the model.

- It estimates a population mean, not the concentration of every individual
- It does not correct for a nonrepresentative sample, measurement error or confounding
- It does not establish that obesity causes lower testosterone levels
- Its strongest scope is a population similar to the study participants

The statistical conclusion should remain aligned with the observational study design.

Example 17.2: Manual Calculation in R

```r
testosterone <- c(
  0.30, 0.24, 0.19, 0.17, 0.18, 0.23, 0.24, 0.06, 0.15,
  0.17, 0.18, 0.17, 0.15, 0.12, 0.25, 0.25, 0.25, 0.32,
  0.35, 0.37, 0.39, 0.46, 0.49, 0.42, 0.36
)
n <- length(testosterone)
x_bar <- mean(testosterone)
s <- sd(testosterone)
se <- s / sqrt(n)
t_star <- qt(0.975, df = n - 1)
margin <- t_star * se
c(lower = x_bar - margin, upper = x_bar + margin)
```

Example 17.2: Confidence Interval with `t.test()`

```
t.test(testosterone, conf.level = 0.95)
```

Relevant output:

```
95 percent confidence interval:
 0.2123626 0.3044374
sample estimates:
mean of x
   0.2584
```

The command also reports a test against the default null value $\mu_0 = 0$. For this example, the confidence interval is the requested result.

What Controls the Width of a t Interval?

$$m = t^* \frac{s}{\sqrt{n}}$$

- A higher confidence level increases t^* and widens the interval
- Greater sample variability increases s and widens the interval
- A larger sample size decreases $s/\sqrt{n}$ and usually narrows the interval
- A different sample can change both $\bar{x}$ and s

Multiplying the sample size by 4 approximately halves the standard error when the underlying variability remains similar.

Common Confidence-Interval Errors

- Using $z^* = 1.960$ when σ is unknown and the sample is small
- Using s instead of $s/\sqrt{n}$ as the standard error
- Forgetting that the procedure uses $df = n - 1$
- Interpreting the interval as a range for individual observations
- Ignoring a severe outlier or a sample-selection problem

Useful Check

The interval should be centred at $\bar{x}$, and its width should reflect the confidence level, variability and sample size.

The One-Sample t Test

To test a claim about an unknown population mean, begin with

$$H_0 : \mu = \mu_0$$

The test statistic is

One-Sample t Statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

with $df = n - 1$.

The statistic measures the distance from the observed sample mean to the null value in estimated standard-error units.

The Alternative Determines the Tail

Let T follow a $t(n-1)$ distribution under H_0.

Alternative	P-**value**
$H_a : \mu > \mu_0$	$P(T \geq t_{\text{obs}})$
$H_a : \mu < \mu_0$	$P(T \leq t_{\text{obs}})$
$H_a : \mu \neq \mu_0$	$2P(T \geq \lvert t_{\text{obs}} \rvert)$

The direction of the alternative should follow from the research question before the sample result is examined.

The Four-Step Hypothesis-Test Process

1. **State:** Define μ and state H_0 and H_a
2. **Plan:** Select a one-sample t test and a significance level if a formal decision is required
3. **Solve:** Check conditions, calculate the test statistic and obtain the P-value from $t(n-1)$
4. **Conclude:** Describe the evidence in context without claiming that H_0 has been proved

Meaning of the P-Value

The P-value is calculated under the assumption that H_0 and the statistical model are true.

Example 17.3: Pulse Wave Velocity and Progeria

Pulse wave velocity (PWV) measures vascular stiffness. A study recorded PWV, in metres per second, for 18 children diagnosed with progeria.

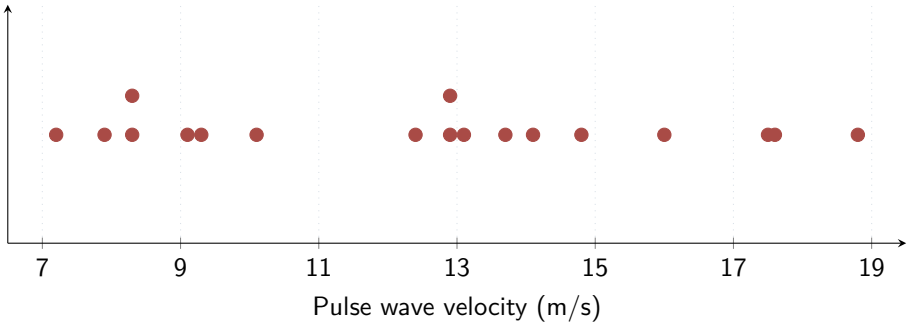
18.8	17.6	17.5	16.0	14.8	14.1	13.7	13.1	12.9
12.9	12.4	10.1	9.3	9.1	8.3	8.3	7.9	7.2

In healthy children, values greater than 6.6 m/s would be considered abnormal.

Question

Do the data provide evidence that the population mean PWV for children with progeria is greater than 6.6 m/s?

Example 17.3: Distribution of PWV



The sample shows no isolated extreme outlier. With $n = 18$ and no severe departure from a single-peaked distribution, a one-sample t test is reasonable.

Example 17.3: Hypotheses

Let μ be the population mean PWV for children with progeria represented by the study.

$$H_0 : \mu = 6.6 \text{ m/s}$$

$$H_a : \mu > 6.6 \text{ m/s}$$

The alternative is upper-tailed because the research question asks whether the mean PWV is higher than the reference value.

Study Scope

Progeria is rare, and the participants were recruited worldwide rather than selected through a formal SRS. The conclusion applies most directly to comparable children with progeria.

Example 17.3: Summary and Standard Error

The sample statistics are

$$n = 18 \qquad \bar{x} = 12.444 \text{ m/s} \qquad s = 3.638 \text{ m/s}$$

The estimated standard error is

$$SE_{\bar{x}} = \frac{3.638}{\sqrt{18}} \approx 0.857 \text{ m/s}$$

If H_0 were true, this standard error would describe the estimated sample-to-sample variability of $\bar{x}$ around 6.6 m/s.

Example 17.3: Test Statistic

The degrees of freedom are

$$df = 18 - 1 = 17$$

The test statistic is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{12.444 - 6.6}{3.638/\sqrt{18}} \approx 6.816$$

The observed sample mean is approximately 6.82 estimated standard errors above the null value.

Example 17.3: P-Value

The alternative is $H_a : \mu > 6.6$, so the P-value is the upper-tail area under the $t(17)$ distribution.

$$P(T \geq 6.816) \approx 0.0000015$$

Interpretation

If the population mean were 6.6 m/s and the model conditions held, a sample mean at least as high as the observed value would occur in approximately 1.5 out of every million repeated samples of size 18.

The very small P-value provides very strong evidence against H_0.

Example 17.3: Conclusion and Effect Size

The data provide very strong evidence that the population mean PWV for comparable children with progeria is greater than 6.6 m/s.

The observed difference from the reference value is

$$12.444 - 6.6 = 5.844 \text{ m/s}$$

Statistical significance addresses evidence against the null model. The size and clinical importance of the difference require subject-matter interpretation and should not be judged from the P-value alone.

Example 17.3: Manual Test in R

```r
pwv <- c(
  18.8, 17.6, 17.5, 16.0, 14.8, 14.1, 13.7, 13.1, 12.9,
  12.9, 12.4, 10.1, 9.3, 9.1, 8.3, 8.3, 7.9, 7.2
)

n <- length(pwv)
x_bar <- mean(pwv)
s <- sd(pwv)
t_stat <- (x_bar - 6.6) / (s / sqrt(n))
p_value <- pt(t_stat, df = n - 1, lower.tail = FALSE)

c(t = t_stat, p_value = p_value)
```

Example 17.3: Test with `t.test()`

```
t.test(
  pwv,
  mu = 6.6,
  alternative = "greater"
)
```

Relevant output:

```
t = 6.8163, df = 17, p-value = 1.501e-06
alternative hypothesis: true mean is greater than 6.6
sample estimates:
mean of x
 12.44444
```

The value supplied to `alternative` must match the stated research hypothesis.

Confidence Intervals and Two-Sided Tests

When a two-sided test and a confidence interval use the same data, model and significance level,

- Reject $H_0 : \mu = \mu_0$ at level α if μ_0 lies outside the corresponding $(1 - \alpha)100\%$ confidence interval
- Fail to reject H_0 if μ_0 lies inside the interval

Different Information

The test evaluates a specific null value. The confidence interval displays a range of plausible population means and communicates the precision of the estimate.

Why Matched Pairs Are Useful

Many studies compare two conditions using observations that are naturally connected.

- The same individual is measured before and after a treatment
- Each participant receives both conditions at different times
- Individuals are paired using relevant characteristics before receiving different treatments

Matching can reduce variation caused by differences among individuals. The analysis must preserve the pairing created by the study design.

Matched Pairs Become One Sample

For each pair, calculate one difference.

$$d_i = \text{response under condition 2} - \text{response under condition 1}$$

The paired dataset becomes

$$d_1, d_2, \ldots, d_n$$

Apply the one-sample t procedures to the differences.

Parameter

μ_d is the population mean of the pairwise differences. A value of $\mu_d = 0$ represents no mean change under the chosen subtraction order.

The Direction of the Difference

The subtraction order determines the sign of the parameter and the alternative hypothesis.

Definition	**Improvement means**	**Alternative**
$d = \text{after} - \text{before}$	Negative differences	$H_a : \mu_d < 0$
$d = \text{before} - \text{after}$	Positive differences	$H_a : \mu_d > 0$

Either definition is valid when it is stated clearly and used consistently throughout the calculation and conclusion.

Matched Pairs Conditions

The one-sample conditions apply to the **differences**.

- The pairs or participants should represent the target population through a defensible sampling process
- The pairwise differences should be independent across pairs
- The distribution of the differences should be approximately Normal, or the number of pairs should be large enough
- Inspect the differences for skewness and outliers

The two measurements within a pair are expected to be dependent. That dependence is the reason for pairing them.

Example 17.4: Treatment for Progeria

The 18 children in the PWV study received lonafarnib for two years. Researchers measured PWV before and after treatment.

Question

Do the paired data provide evidence that the treatment lowers mean PWV among comparable children with progeria?

Define

$$d = \text{treated PWV} - \text{untreated PWV}$$

A reduction produces a negative difference. The hypotheses are

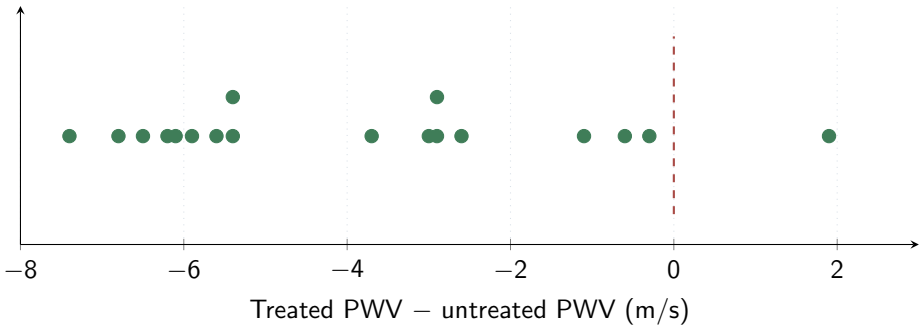
$$H_0 : \mu_d = 0 \qquad H_a : \mu_d < 0$$

Example 17.4: Paired Measurements

Child	Before	After	d	Child	Before	After	d
1	18.8	12.6	−6.2	10	12.9	7.5	−5.4
2	17.6	10.8	−6.8	11	12.4	9.4	−3.0
3	17.5	10.1	−7.4	12	10.1	6.4	−3.7
4	16.0	10.1	−5.9	13	9.3	9.0	−0.3
5	14.8	9.2	−5.6	14	9.1	6.2	−2.9
6	14.1	7.6	−6.5	15	8.3	7.2	−1.1
7	13.7	10.8	−2.9	16	7.9	7.3	−0.6
8	13.1	7.7	−5.4	17	8.3	5.7	−2.6
9	12.9	6.8	−6.1	18	7.2	9.1	1.9

The measurements are paired by child. The 18 differences, not the 36 individual measurements, form the sample for the t analysis.

Example 17.4: Distribution of Differences



The distribution shows mild skewness and no isolated extreme outlier. A paired t procedure is reasonable for the 18 differences.

Example 17.4: Summary Statistics

For the 18 differences,

$$n = 18 \qquad \bar{d} = -3.917 \text{ m/s} \qquad s_d = 2.644 \text{ m/s}$$

The standard error is

$$SE_{\bar{d}} = \frac{s_d}{\sqrt{n}} = \frac{2.644}{\sqrt{18}} \approx 0.623 \text{ m/s}$$

The negative sample mean indicates that PWV was lower after treatment, on average, in the observed sample.

Example 17.4: Test Statistic and P-Value

The test statistic is

$$t = \frac{\bar{d} - 0}{s_d/\sqrt{n}} = \frac{-3.917}{2.644/\sqrt{18}} \approx -6.286$$

with $df = 17$. Because $H_a : \mu_d < 0$, the P-value is the lower-tail area.

$$P(T \leq -6.286) \approx 0.0000041$$

The paired data provide very strong evidence that the treatment lowers mean PWV for comparable children with progeria.

Example 17.4: Confidence Interval for Mean Change

A 95% interval for the mean difference is

$$\bar{d} \pm t^* \frac{s_d}{\sqrt{n}}$$

$$-3.917 \pm 2.110(0.623)$$

$$(-5.231,\ -2.602) \text{ m/s}$$

Using $d = \text{after} - \text{before}$, the interval is negative. Equivalently, the estimated population mean reduction is between approximately 2.60 and 5.23 m/s.

Statistical and Clinical Importance

The very small P-value provides strong evidence of a mean reduction in PWV. The confidence interval estimates the size of that reduction.

Neither result determines whether the reduction is clinically important. That judgment requires knowledge of

- How a change in PWV relates to health outcomes
- The treatment's side effects and burdens
- The variation in responses among individual patients

Statistical evidence should be combined with clinical expertise and the full study design.

Example 17.4: Paired Test in R

```r
before <- c(
  18.8, 17.6, 17.5, 16.0, 14.8, 14.1, 13.7, 13.1, 12.9,
  12.9, 12.4, 10.1, 9.3, 9.1, 8.3, 7.9, 8.3, 7.2
)
after <- c(
  12.6, 10.8, 10.1, 10.1, 9.2, 7.6, 10.8, 7.7, 6.8,
  7.5, 9.4, 6.4, 9.0, 6.2, 7.2, 7.3, 5.7, 9.1
)

t.test(after, before, paired = TRUE, alternative = "less")
```

R forms `after - before`, matching the stated definition of the differences.

Why the Pairing Must Be Preserved

Treating the before and after measurements as two independent samples would ignore the connection between observations from the same child.

- The independent-samples model assumes that the two groups do not share observational units
- The matched pairs model uses each child as their own comparison
- The variation among children is partly removed by working with within-child differences

General Principle

The inferential procedure must reflect the design that produced the data.

Robust Statistical Procedures

A procedure is **robust** when moderate violations of its conditions have little effect on the reported confidence level or P-value.

One-sample t procedures are reasonably robust to non-Normal population shapes when the sample is large enough.

Important Exception

The procedures are sensitive to outliers because both $\bar{x}$ and s can change substantially when one extreme observation is present.

Robustness to distribution shape does not correct selection bias, dependence or measurement problems.

Practical Sample-Size Guidelines

Sample size	Practical guidance
$n < 15$	Use t procedures when the data are roughly symmetric and single-peaked, with no outliers
$15 \leq n < 40$	Mild skewness is usually acceptable, but strong skewness or outliers remain concerning
$n \geq 40$	Strong skewness is often manageable, although influential outliers still require investigation

These are guidelines rather than automatic rules. The data-collection design remains more important than the exact population shape.

A Practical Screening Process

Before using a one-sample or matched pairs t procedure:

1. Identify the observational units and the target population
2. Confirm whether the observations or pairwise differences can be treated as independent
3. Plot the observations or differences
4. Look for isolated outliers, strong skewness and multiple groups
5. Consider the sample size together with the observed shape
6. Investigate unusual values before removing or altering them

The graph and study design determine whether the numerical procedure answers a meaningful question.

Example 17.5: Small and Moderate Samples

Dataset	Size	Assessment
Oak acorn volumes	$n = 11$	One clear, extreme outlier makes t procedures unreliable for this small sample
Aleppo pine needle lengths	$n = 15$	Mild right skewness with no outliers is acceptable for a moderate sample

The difference between the examples is not only the sample size. The extreme outlier in the acorn data directly affects both the sample mean and sample standard deviation.

Example 17.5: Large Samples

Dataset	Size	Assessment
Brown treesnake weights	About 500	Strong right skewness is manageable at this size, but the heaviest snake should still be investigated for influence
Wild bee cocoon widths	About 800	Computation is stable, but the clear bimodality makes one overall mean scientifically difficult to interpret

A large sample can improve the sampling distribution of $\bar{x}$. It cannot make a scientifically unhelpful summary meaningful.

Bimodality and the Meaning of a Mean

The wild bee cocoon distribution contains two groups because female and male cocoons differ in width.

An overall mean combines

- The mean width within each biological group
- The proportion of observations belonging to each group

If the sex composition changes, the overall mean can change even when neither group changes.

Analysis Choice

Separate summaries or a model that includes sex may answer the biological question more clearly than one confidence interval for a pooled mean.

When an Outlier Appears

An unusual observation should be examined rather than deleted automatically.

- Check for data-entry, unit or measurement errors
- Confirm whether the observation belongs to the target population
- Compare the analysis with and without the observation as a sensitivity check
- Report any decision to exclude an observation and explain the reason
- Consider a transformation or another method when the population is strongly skewed

An outlier may be an error, a valid rare value or evidence that the population contains distinct subgroups.

Choosing the Correct Procedure

Study structure	**Procedure**
One quantitative sample, unknown σ	One-sample t interval or test
Two measurements on each individual	Paired t procedure applied to within-individual differences
Legitimately matched pairs	Paired t procedure applied to within-pair differences
Two separate, unrelated groups	Independent two-sample procedure, studied in the next chapter

The number of columns in a dataset does not determine the method. The design and dependence structure do.

Chapter 17 Summary

- When σ is unknown, estimate the standard error of $\bar{x}$ with $s/\sqrt{n}$
- Use the $t(n-1)$ distribution for one-sample confidence intervals and tests
- A level C confidence interval is $\bar{x} \pm t^* s/\sqrt{n}$
- A test of $H_0 : \mu = \mu_0$ uses $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$
- Matched pairs procedures apply one-sample t methods to pairwise differences
- Inspect the data and study design before relying on the procedure

Final Principle

The calculation quantifies sampling uncertainty. The design, data quality and scientific context determine what the result can mean.